

Roudy Hajo

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Final-year Computer Science Engineering student specializing in Networks, Cloud Computing, and Security at Polytech Nantes. Passionate about software development, distributed systems, and cloud infrastructure. Seeking a 6-month Software Engineer internship starting February 2027.

EDUCATION

Polytech Nantes School of Engineering <i>Master's Degree in Computer Science (Networks, Security, Cloud Computing)</i>	Nantes, France Sep. 2024 – Aug. 2027
University Institute of Technology of Nantes <i>Technical Associate's Degree in Computer Science</i>	Nantes, France Sep. 2022 – Jun. 2024

EXPERIENCE

Research Intern – Distributed DBMS Indexing <i>Coimbra Institute of Engineering</i>	May 2026 – Aug. 2026 Coimbra, Portugal
- Designed a decentralized multi-agent QNN and reinforcement learning architecture to optimize database index tuning. - Engineered an adaptive centralized learning router to eliminate workload skew and balance queries across replicas.	
Web Developer Intern <i>Nantes University – DSIN</i>	May 2025 – July 2025 Nantes, France
Optimized the 'Syllabus' Symfony web application, refactoring core modules to dynamically present course offerings.	
Web Developer Intern <i>Kookline – Web Development Agency</i>	Apr. 2024 – June 2024 Nantes, France
Developed a custom WordPress web platform automating skydiving reservations and scheduling workflows.	
Academic Tutor <i>University Institute of Technology of Nantes</i>	Oct. 2023 – July 2024 Nantes, France
Mentored 50+ first-year students in core computer science, focusing on algorithms, OOP, data structures, and mathematics.	

PROJECTS

I2PoFM (Invisible Internet over FM) <i>Python, GNU Radio, I2P, SDR, Raspberry Pi</i>	Feb. 2026 – Mar. 2026
- Engineered a stealth communication system to multiplex encrypted IP traffic within standard FM radio broadcasts using GNU Radio, DSSS, and PCM techniques for complete signal inaudibility. - Integrated the decentralized I2P network layer onto Raspberry Pi hardware to route network packets covertly, achieving high data stealth, anonymity, and confidentiality.	
SketchOScent <i>Python, Webpack, JavaScript, RDKit, PubChem</i>	Sep. 2025 – May 2026
- Enhanced a cheminformatics web application mapping chemical functional groups to specific scent profiles using Python. - Integrated an automated code obfuscation pipeline into Webpack to effectively secure intellectual property.	
PolyRisk <i>Python</i>	Oct. 2025 – Nov. 2025
Developed intelligent AI agents for the Risk board game utilizing custom heuristics and Monte Carlo Tree Search (MCTS).	
Dice Wars <i>C++</i>	Nov. 2025 – Jan. 2026
Engineered a turn-based strategy game featuring heuristic-driven AI opponents and procedural map generation.	
NDAROS <i>Python, Arduino, HTML/CSS, JavaScript</i>	Feb. 2025 – Mar. 2026
Designed an interactive physical model with real-time Arduino algorithms demonstrating cosmic energy production.	
Football App <i>JS, Node.js, Express.js, Passport.js, React, Tailwind CSS</i>	Mar 2024 – Apr 2024
Built a full-stack platform with external REST API caching and an authenticated multi-user betting game.	

TECHNICAL SKILLS

Languages: Go, Java, Kotlin, Python, C/C++, SQL (Oracle, MariaDB), MongoDB, HTML/CSS, JavaScript, R
Frameworks: Spring Boot, Angular, React, Node.js, Express.js, Symfony, Flask, JUnit
Developer Tools: Git, Docker, Kubernetes, Microsoft Azure, VS Code, Visual Studio, IntelliJ
Languages Spoken: French (Native), English (Professional Proficiency), Arabic (Fluent), Kurdish (Native), German (A1)

INTERESTS

Hobbies & Activities: Futsal, Board games, Open-Source Innovation, Traveling, Reading, Global current events